

Which sample drains most?

Learning intention

Compare how much water passes through different soils in the same time.

A puddle stays on one patch.

Another patch drains sooner.

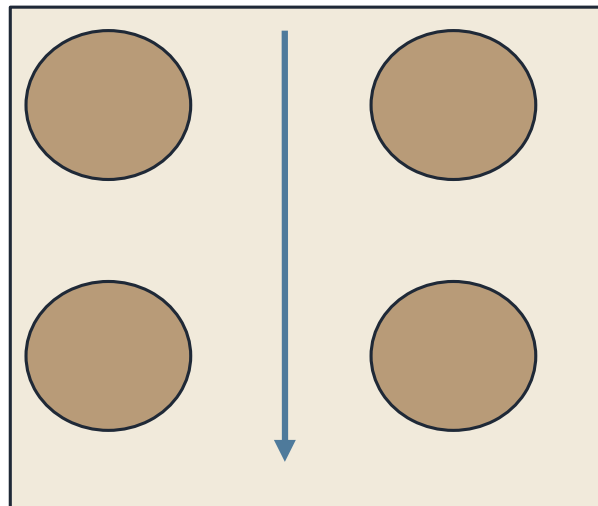
How can we compare soils fairly?

Find the missing links

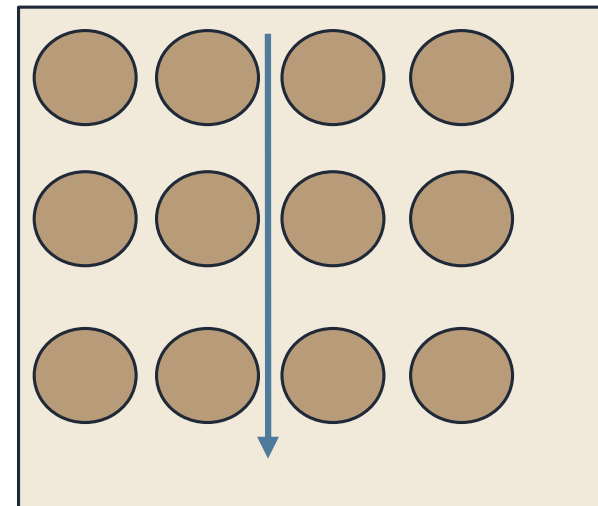
1. Name two solid parts of soil.
2. What is a pore?
3. Which is more: 20 mL or 40 mL?

Follow a connected path

Broad connected paths



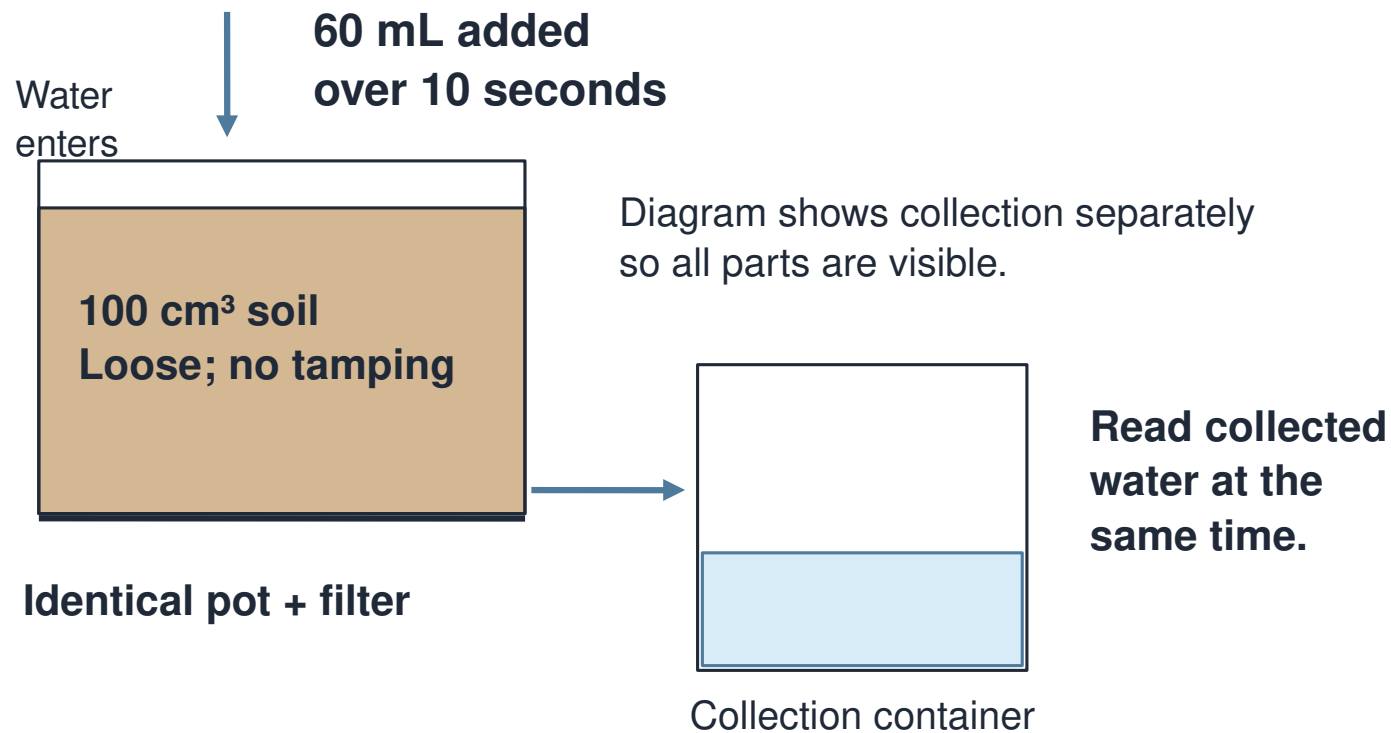
Narrow connected paths



A model of pathways, not a rule for every soil.

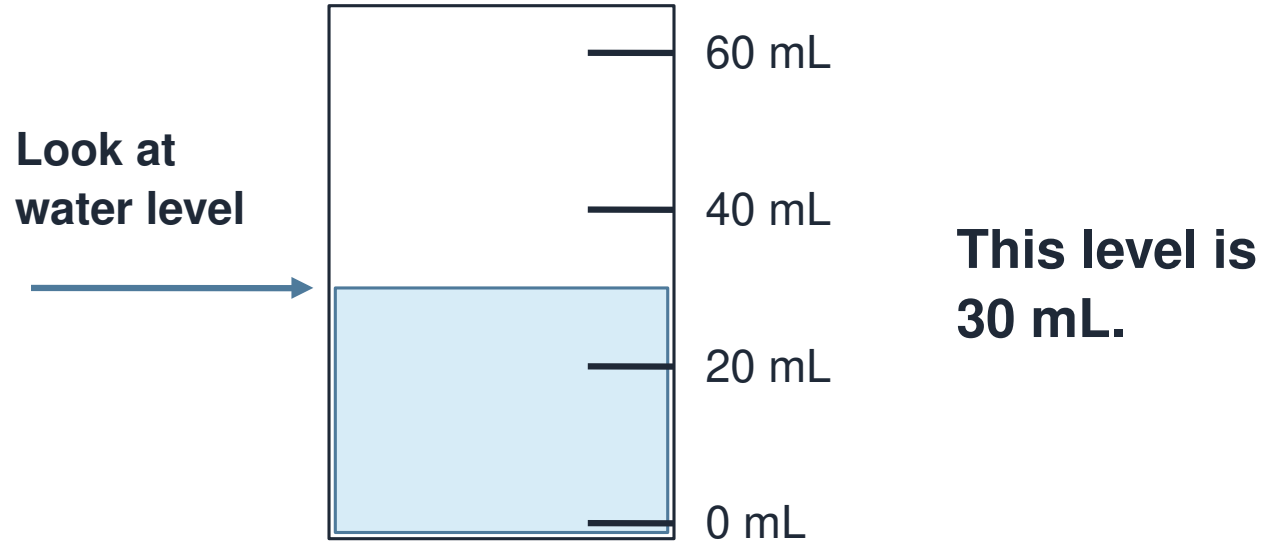
Water moves through connected pores. Pore size and connections matter. Real soil is more complex.

Give each soil the same test



Same soil volume: 100 cm³. Same water: 60 mL. Read at exactly 2 minutes.

Read the scale carefully



Read between marks. Keep the container upright.

Keep the container upright. Look at the water level. Read the volume in mL. What volume is shown?

Repair the unfair test

Choose the useful test cards. Fix any card that changes two things.

Card	Plan
A	All pots: 60 mL; read at 2 min
B	A: 60 mL; B: 100 mL
C	All pots: 100 cm ³ ; loose fill
D	Pack C tightly; leave A and B loose

Constructed planning cards; identical apparatus is assumed

Read; compare; justify

Compare trial 1. Which sample let most water through? What makes this a fairer comparison?

Soil	Trial 1: mL out	Trial 2: mL out
A	42	40
B	26	28
C	10	12

Constructed sample data • 60 mL in • collect at 2 minutes

What can we conclude?

Choose the best claim.

A A let most water through in this two-minute test.

B C is the best soil for every plant.

C A must always drain fastest outdoors.

Take a role; choose a route

Timer, reader or recorder.

Or work with the paper data.

Use a result to justify your claim.

Our method in six actions

Fill; measure; start timer.

Pour; collect; read.

Keep the conditions the same.

Make the evidence count

Write the volume with mL.

Compare A, B and C.

Finish: “In this test, ____ because ____.”

Check before you claim

Did each test use the same input?

Did we read at the same time?

Does my claim name this test?

Lundy loop: improve our test

Share one result or concern.

Listen for a different observation.

Choose one useful improvement.

What would make our comparison more trustworthy?

Leave with an evidence claim

E1. Which drained most in the sample data?

E2. Name one condition kept the same.

E3. Can this choose the best soil for every plant?